

conditions into a single central control area, wherein a computer would calculate the weather forecast for that cell. On completion of all calculations, cell-wise data would be posted and the process would restart.

In this way, 64,000 people processing a constantly-evolving set

of numbers would be able to arrive at non-linear equations essential to forecast the planet's weather.

This method gave birth to two technologies that are currently in use, namely, 64,000-microprocessor supercomputers used by the government and large enterprises and computer-controlled mobile communication grids similar to those used in mobile phone networks.

It was John Mauchly, a computer scientist, who pioneered the usage of Electronic Numerical Integrator and Computer, or ENIAC, the first electronic general-purpose computer, as a tool to facilitate long-term weather prediction.



Fig. 2: Staff programming ENIAC at University of Pennsylvania, USA, circa 1946 (Source: www.fortune.com)

How weather maps are created

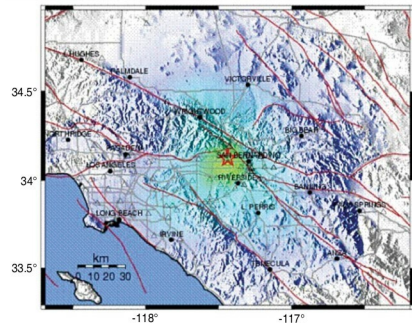
Using various gadgets, instruments and tools, meteorologists state the present conditions of the atmosphere. These are then plotted on weather maps by computers to showcase the complete condition of the atmosphere around the globe. Quality control is achieved by computers, which rectify possible errors in the data related to weather. This leads to higher accuracy and reliability in reports.

To begin with, advanced computing systems are fed data to draw independent weather maps for the seven levels of the atmosphere used in forecasting. Once data has been analysed in the form of maps, IT is used to provide inputs by way of drawn maps, to determine the weather in each of the seven-layer grid. Thereafter, processing is carried out, that is, the various number crunching and tracking of the dynamic weather conditions. Final output encompasses the different types of weather maps based on a number of data points that have been collected and analysed.

Numerical forecasting is a term that refers to the usage of mathematical equations to forecast the weather and is based on a large number of very-high-speed calculations that only supercomputers are capable of performing. It is inferred that these equations are more multifaceted than those used in aerospace engineering.

On an average, National Weather Service (NWS) receives as high as 100 million weather observations every day. Data on a number of different variables such as wind speed, air temperature, barometric pressure and humidity is collected from many sources including land based observation points and ships moving around in the ocean. This information is then fed into a supercomputer, which usually has two sets of several fast processors capable of parallel processing. The

CISN Rapid Instrumental Intensity Map Epicenter: 1.3 mi N of Fontana, CA
Thu Jan 6, 2005 06:35:27 AM PST M4.4 N34.12 W117.44 Depth: 4.2km ID: 14116972



Processed: Wed Jan 12, 2005 06:29:32 PM PST.—NOT REVIEWED BY HUMAN

PERCEIVED SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
POTENTIAL DAMAGE	None	None	None	Very light	Light	Moderate	Moderate/Heavy	Heavy	Very heavy
PEAK ACC (%g)	<0.1	0.1-1.4	1.4-3.9	3.9-9.2	9.2-18	18-34	34-65	65-124	>124
PEAK VEL (cm/s)	<0.1	0.1-1.1	1.1-3.4	3.4-8.1	8.1-16	16-31	31-60	60-116	>116
INSTRUMENTAL INTENSITY	I	II-III	IV	V	VI	VII	VIII	IX	X+

Fig. 3: A computer-generated shake map (Source: www.trinet.org)